ACADEMIC CENTRAL

PRACTICE EXAMINATION

N6

CONTROL SYSTEMS N6

PRACTICE EXAM

MARKS: 100 **TIME: 3 hours**

NATED Engineering Studies, N6. Block diagrams and transfer functions, transient response of second-order systems, steady-state error, stability (Routh–Hurwitz), controller actions (P, I, D and on–off). A formula sheet is attached.

Written by Academic Central for practice, following the NATED (Report 191) syllabus for N6. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations. Write units with all answers.
5. Sketches and block diagrams must be neat, large and fully labelled.
6. A formula sheet is attached at the end of this question paper.
7. Round off answers to THREE decimal places, unless stated otherwise.
8. Write neatly and legibly.

QUESTION 1: BLOCK DIAGRAMS AND TRANSFER FUNCTIONS

1.1 Draw the block diagram of a closed-loop control system with negative feedback. Label the reference input, error signal, forward path G(s), feedback path H(s) and output. (4)

1.2 Derive the closed-loop transfer function $\frac{C(s)}{R(s)}$ of the system in QUESTION 1.1. (4)

1.3 A unity-feedback system has an open-loop transfer function $G(s) = \frac{10}{s(s+2)}$. Determine the closed-loop transfer function. (3)

1.4 Two blocks, $G_1 = 2$ and $G_2 = \frac{5}{s+1}$, are connected in cascade (series) in the forward path, with a feedback block H = 0,1.

1.4.1 Determine the overall transfer function. (5)

1.4.2 Calculate the final value of the output for a unit step input. (4)

[20]

QUESTION 2: TRANSIENT RESPONSE

2.1 For the closed-loop system in QUESTION 1.3, calculate:

2.1.1 the undamped natural frequency (2)

2.1.2 the damping ratio, and classify the damping (3)

2.1.3 the damped natural frequency (2)

2.1.4 the percentage overshoot (3)

2.1.5 the peak time (2)

2.1.6 the settling time (2% criterion) (2)

2.2 Sketch, on the same axes, the unit step responses of a second-order system that is underdamped, critically damped and overdamped. (6)

[20]

QUESTION 3: STEADY-STATE ERROR

3.1 Explain what is meant by the "type" of a control system. (2)

3.2 A unity-feedback system has $G(s) = \frac{20}{(s + 2)(s + 5)}$.

3.2.1 State the type of the system. (1)

3.2.2 Calculate the position error constant and the steady-state error for a unit step input. (4)

3.3 For the system in QUESTION 1.3, calculate the steady-state error for (a) a unit step input and (b) a unit ramp input. (6)

3.4 Explain how adding integral action affects the steady-state error. (2)

[15]

QUESTION 4: STABILITY

4.1 A system has the characteristic equation $s^3 + 6s^2 + 11s + K = 0$.

4.1.1 Construct the Routh array. (5)

4.1.2 Determine the range of values of K for which the system is stable. (4)

4.1.3 Calculate the frequency of sustained oscillation when K is at its critical value. (3)

4.2 Explain what is meant by a stable system, in terms of the response to a bounded input and in terms of the position of the poles in the *s*-plane. (4)

4.3 Give TWO effects of increasing the gain of a closed-loop system. (4)

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QUESTION 5: CONTROLLERS

5.1 Describe the action of each of the following controllers and sketch its output for a step change in the error: (a) proportional (b) integral (c) derivative (9)

5.2 A proportional controller has a proportional band of 20%. Calculate the controller gain. (2)

5.3 Explain why derivative action is never used on its own. (2)

5.4 An oven is controlled by an on–off (two-position) controller with a set point of 180 °C and a differential gap of ±3 °C.

5.4.1 Sketch the oven temperature and the heater state against time. (4)

5.4.2 Explain why a differential gap (hysteresis) is used. (2)

5.5 Name TWO methods of tuning a PID controller. (2)

5.6 Name and describe FOUR elements of an industrial process control loop (e.g. a level-control system). (4)

[25]

TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

Closed loop: $\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$

Final value theorem: $c(\infty) = \lim_{s\to 0} sC(s)$

Second-order: $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$

$\%OS = e^{-\zeta\pi/\sqrt{(1-\zeta^2)}} \times 100$

$t_p = \frac{\pi}{\omega_d}$; $t_s\ (2\%) = \frac{4}{\zeta\omega_n}$

$K_p = \lim_{s\to 0} G(s)$; $e_{ss(step)} = \frac{1}{1 + K_p}$

$K_v = \lim_{s\to 0} sG(s)$; $e_{ss(ramp)} = \frac{1}{K_v}$

Gain $= \frac{100}{PB(\%)}$